FINAL REPORT

E-Commerce Sales, Demand Forecasting & Customer Churn Analysis

Sales analysis, demand forecasting and customer churn with real Brazilian e-commerce data, as a foundation for an agricultural machinery business

The four business questions

1. Which products and categories sell the most and the least?
2. How much will sell in the next 3 months, and how much inventory is needed?
3. Where do customers come from, and where should a new branch open?
4. Why do some customers not come back, and who is about to leave?

Data: Brazilian E-Commerce Public Dataset by Olist, 2016 to 2018
Tools: SQL (SQLite), Python (pandas, statsmodels, scikit-learn, SHAP), Power BI
October 2026

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1. Executive summary

This report answers four business questions for an agricultural machinery business whose sales system cannot export its data yet. The analysis uses the public Olist dataset, about 100,000 real e-commerce orders from Brazil with the same structure (products, sale dates, customers and locations), and every step is ready to be repeated with the business's own data.

Question	What we found	What we recommend
1. Which products and categories sell the most and the least?	25.9% of the products bring in 80% of revenue, while 6,707 slow movers (20.9% of the catalog) bring in only 2.6%.	Guarantee the availability of the 8,309 class A products and stop restocking the slow movers.
2. How much will sell in the next 3 months, and how much inventory is needed?	About 7,259 units per month from August to October 2018. A 3-month moving average (7.8% error) beat Holt and ARIMA; at category level the error is around 24%.	Plan every month with the moving average, keep a safety stock per category and order items with intermittent demand on request.
3. Where do customers come from, and where should a new branch open?	41.9% of the customers are in the state of São Paulo. Bahia is the most robust candidate for a branch, and Salvador the best-supported city.	Evaluate a branch in Salvador after agreeing how much demand and service quality should weigh.
4. Why do some customers not come back, and who is about to leave?	98.25% of the customers do not come back within 180 days. The first purchase predicts little (ROC AUC 0.571), but coming back is associated with the region, everyday categories and the first purchase experience.	Build a post-purchase retention program and test every action with a control group.

Three ideas run through the answers. First, simple methods held up better than complex ones: a moving average beat the trend and autoregressive models, and a weak churn model is presented as weak. Second, the Northeast appears in two answers: it is the best region for a new branch, and its customers come back less than those of the Southeast, which points to delivery service as a common lever. Third, every result is an association measured in a proxy business, so the recommendations are hypotheses to confirm with the machinery business's data.

2. Business problem and approach

An agricultural machinery business needs to decide which items to promote or reduce, how much stock to keep, where to open a new branch and how to retain its customers. Its sales system does not allow exporting the data without paying for a more expensive plan, so the analysis was developed with a public dataset of the same structure and documented to be repeated as soon as the business's data is available.

From data to decisions

The work follows the flow **SQL → Python → models → Power BI**. The first notebook cleans the raw files and loads the clean tables into a SQLite database. From then on, aggregations are written in SQL, one query per file, and Python takes care of statistics, charts and models. The results are saved as CSV files for the dashboard.

Phase	Notebook	What it does
1	`01_data_cleaning`	Joins and cleans the tables, computes revenue, delivery time and delay, and creates the database
2	`02_exploratory_analysis`	Sales by month, day and hour; seasons; ticket; categories; regions
3	`03_products`	Rankings, Pareto analysis, ABC classification, slow movers, category trends
4	`04_forecast_inventory`	Baselines against Holt and ARIMA, 3-month forecast, safety stock
5	`05_geographic`	Customer map, service gap by state, branch ranking and sensitivity analysis
6	`06_customer_churn`	Churn definition, leakage checks, rates with confidence intervals, two models and SHAP
7	Power BI	One dashboard page per business question

How each question is answered

Every question closes with the same structure: **Business question → Analysis → Evidence → Insight → Recommendation → Application to the machinery business**, followed by its limitations. The last step matters because Olist is a proxy: each recommendation is translated into what it means for a business that sells tractors, implements and spare parts. Criteria and decision rules, such as the weights of the branch ranking or the rule for choosing the forecasting model, were written before looking at the results.

3. Data and cleaning

The **Brazilian E-Commerce Public Dataset by Olist** (CC BY-NC-SA 4.0 license) contains about 100,000 orders placed between 2016 and 2018, spread across 9 CSV files: orders, items, products, customers, geolocation, reviews, payments, sellers and a category translation table.

Cleaning keeps the 96,470 delivered orders that have a delivery date, out of 99,441. Four traps in the dataset would silently inflate the numbers when joining tables: Olist creates a new `customer_id` for every order, an order can have several payments, some orders have several reviews, and the geolocation table repeats each zip code prefix many times. Every join is validated so that it cannot duplicate rows, and every step that removes or summarizes rows is recorded in a log:

Table	Step	Rows before	Rows after	Reason
Orders	Delivered orders only	99,441	96,478	Canceled, in transit or not shipped: not completed sales yet
Orders	Delivered without delivery date	96,478	96,470	Delivery time and delay cannot be measured without a delivery date
Payments	One row per order	103,886	99,440	Payments of the same order are added up so amounts are not duplicated when joining
Reviews	One review per order	99,224	98,673	Orders with several reviews keep the most recent answer
Geolocation	Points outside Brazil	1,000,163	1,000,121	Impossible coordinates for Brazilian addresses: data entry errors
Geolocation	One point per zip code prefix	1,000,121	19,010	Median latitude and longitude of each prefix

Table	Step	Rows before	Rows after	Reason
Items	Items from delivered orders only	112,650	110,189	Items from orders that were not delivered are not sales

Data coverage

2016 has only 267 delivered orders, and from August 22, 2018 the number of orders drops abruptly in every order status: it is the end of the available data, not a fall in sales. That is why the analysis uses January 2017 to August 2018, with 19 complete months (January 2017 to July 2018) for the forecast and daily averages computed up to August 21, 2018.

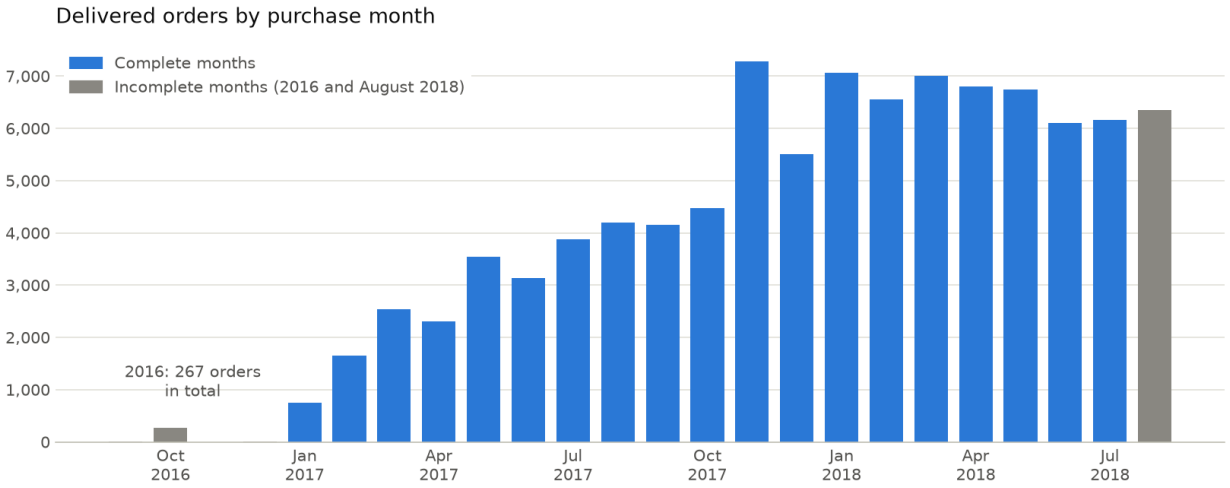


Figure 1. Delivered orders by purchase month. Gray months are incomplete: 2016 and August 2018, whose data stops being complete on August 22.

4. Sales context

Before answering the business questions, the exploratory analysis describes how sales behave:

- **Growth came from volume, not price.** Orders grew 160.8% between January to July 2017 and January to July 2018, while the average ticket stayed between R$ 124 and R$ 149.
- **Two stages.** Sales grew almost every month of 2017 and stabilized in 2018 at 6,100 to 7,100 orders per month.
- **Black Friday is the clearest seasonal effect.** November 24, 2017 had 1,147 orders, against a median of 183 orders per day that month.
- **Small, single-item purchases.** The median order is R$ 86.50, and 90.0% of the orders contain a single unit.
- **Weekdays and daytime.** Weekend sales are 25.3% lower than on weekdays, and 81.5% of the orders are placed between 10:00 and 22:59.
- **Concentration in the Southeast.** The region brings in 65.4% of revenue and hosts 73.7% of the sellers.

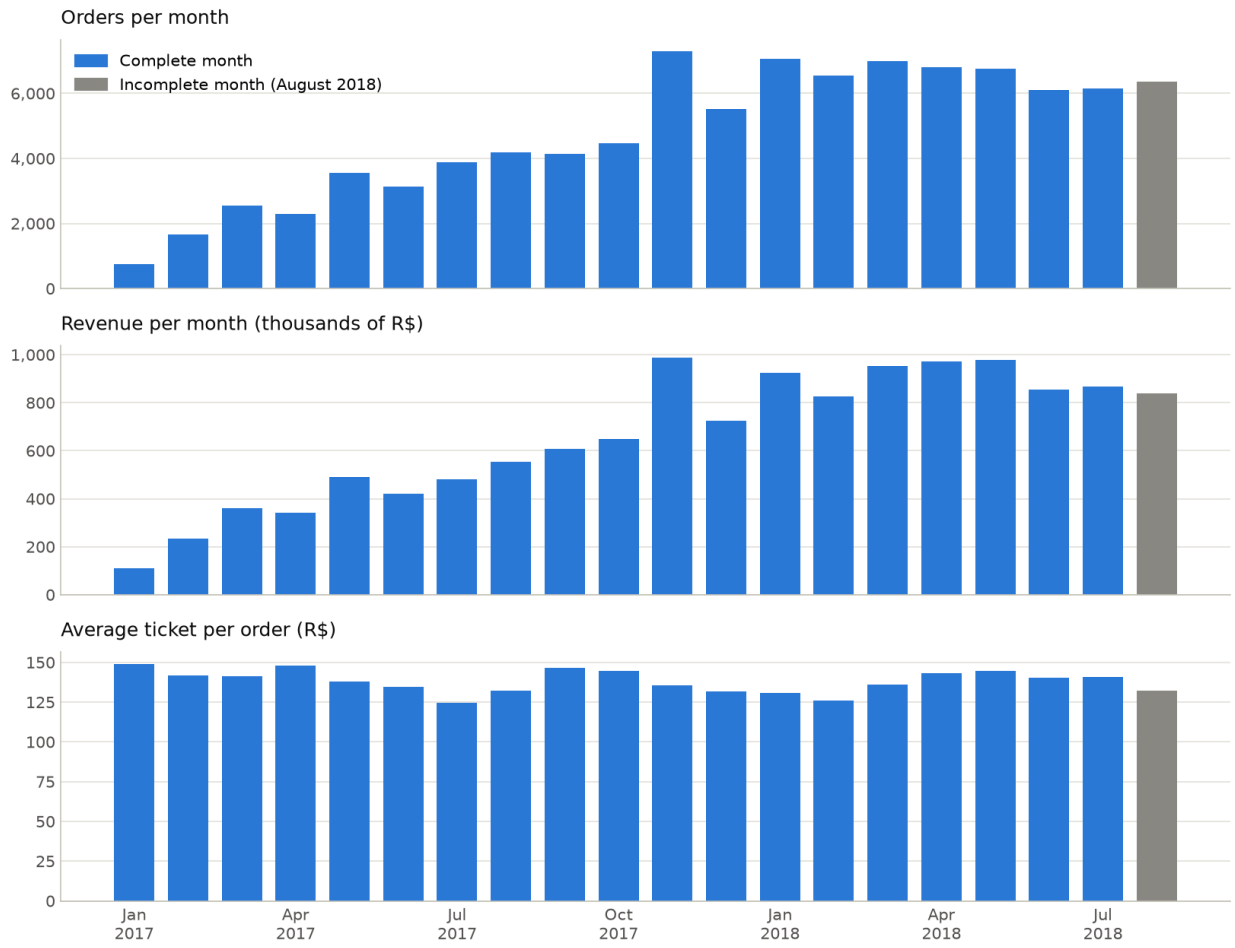


Figure 2. *Orders, revenue and average ticket per month. Orders and revenue grow together while the ticket stays flat: growth came from the number of orders.*

5. Question 1 · Products that sell the most and the least

We ranked the 74 categories and the 32,079 products sold between January 2017 and August 2018, measured revenue concentration and classified each product as A (first 80% of revenue), B (next 15%) or C (last 5%).

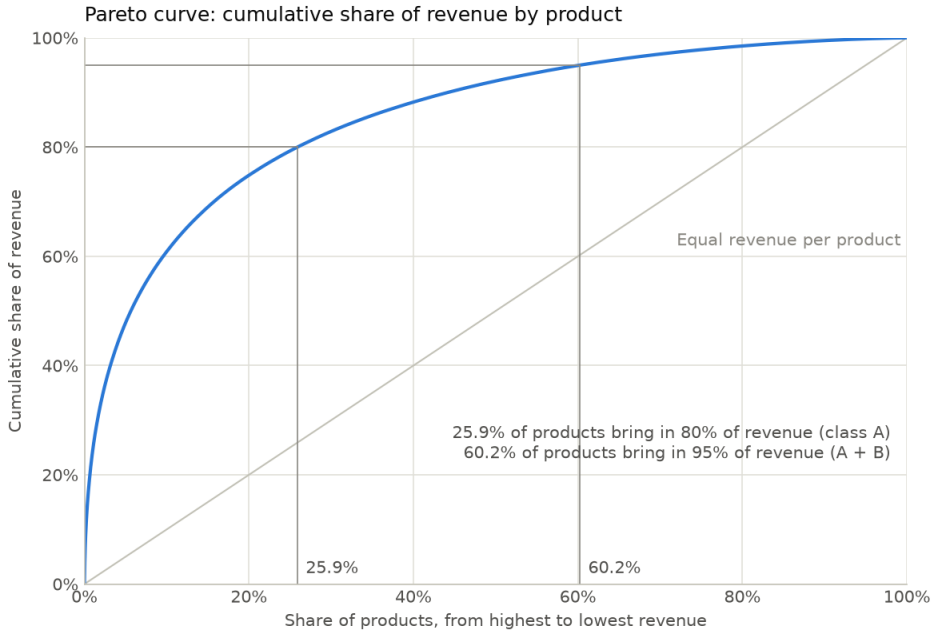


Figure 3. *Pareto curve. 25.9% of the products bring in 80% of revenue, and 60.2% bring in 95%.*

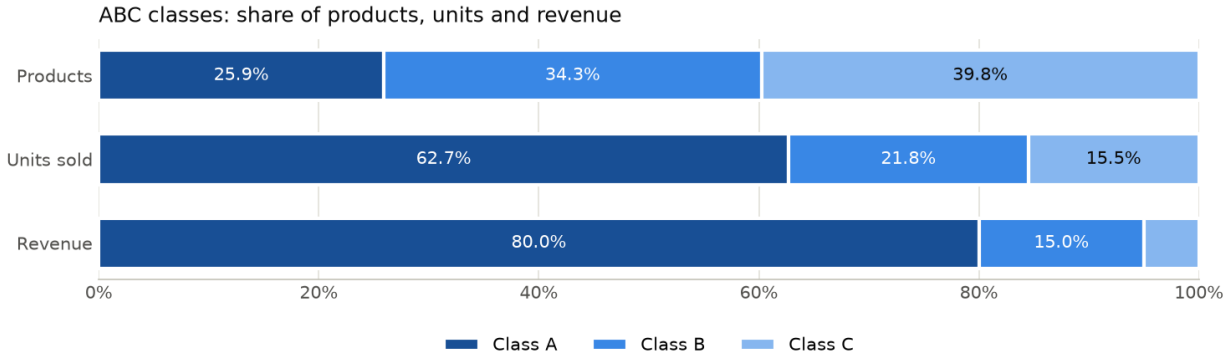


Figure 4. *ABC classes. Class A is a quarter of the products, but it concentrates 62.7% of the units and 80% of revenue.*

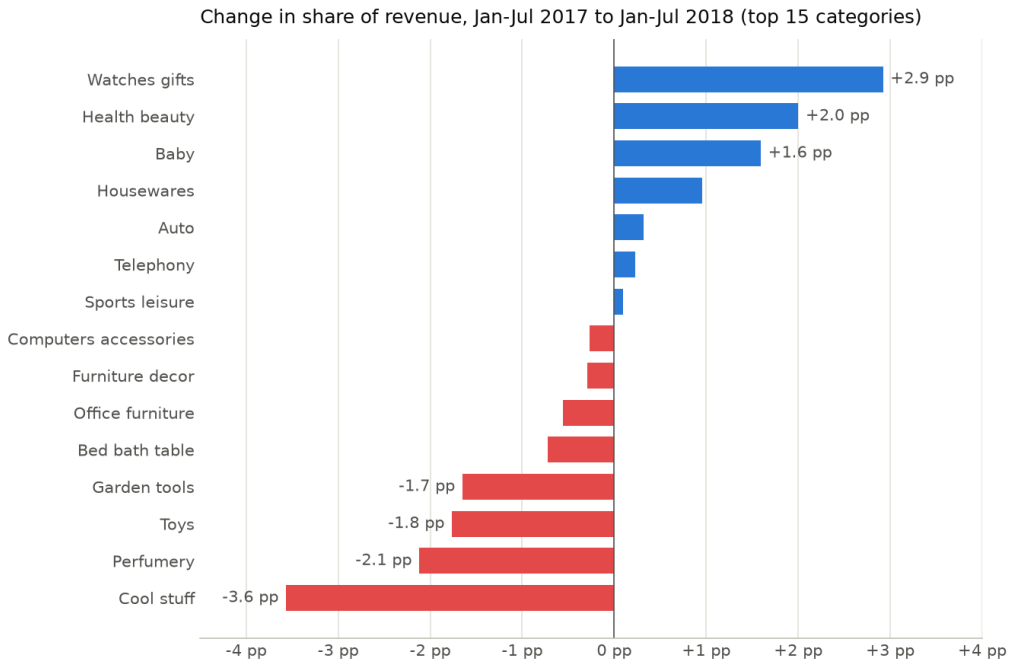


Figure 5. *Change in each category's share of revenue between January to July 2017 and January to July 2018 (pp = percentage points).*

Answer to question 1

Business question. Which products and categories sell the most and the least?

Analysis. We ranked the 74 categories and the 32,079 products sold between January 2017 and August 2018 by revenue and by units, measured revenue concentration with a Pareto curve, classified each product as A, B or C, identified slow movers and compared each category's share of revenue between the first seven months of 2017 and of 2018.

Evidence.

- 25.9% of the products (class A) bring in 80% of revenue, while 39.8% (class C) bring in only 5%.
- 55.0% of the products sold a single unit in 20 months, and 6,707 products (20.9%) have not sold since February 22, 2018.
- 18 of the 74 categories bring in 80% of revenue, led by Health beauty, Watches gifts and Bed bath table.
- The ranking by units differs from the ranking by revenue: Bed bath table sells the most units but ranks third in revenue.
- Watches gifts (+2.9 pp), Health beauty (+2.0 pp) and Baby (+1.6 pp) gained share, while Cool stuff (−3.6 pp), Perfumery (−2.1 pp) and Toys (−1.8 pp) lost it.

Insight. Revenue depends on a relatively small group of products, while a long tail of rarely sold products brings in very little. Volume and revenue do not always go together, and categories are moving: some gain ground steadily, others decline, and at least one (Toys) appears to depend on the Christmas season.

Recommendation.

1. Treat the 8,309 class A products as the priority: they should never run out of stock, and they are the natural candidates for promotions and visibility.
2. Review the 6,707 slow movers: stop restocking them, sell them on demand, bundle them with A products or remove them from the catalog.
3. Expand the assortment in growing categories (Watches gifts, Health beauty, Baby), and investigate why Cool stuff and Perfumery are declining (prices, competition, seller issues) before cutting them.
4. Plan seasonal categories such as Toys around their peak months.

Application to the machinery business. The products notebook can classify the business's items as soon as its sales data can be exported. We expect a few high-value items (tractors and implements) to fall in class A and many spare parts to spread across B and C, but real data has to confirm it. In practice:

- Class A items need guaranteed availability and close follow-up of stock and supplier lead times.
- Class C items and slow movers should be kept at minimal stock or ordered on demand. Machinery stock that does not move ties up a lot of money, so the six-month rule is especially useful there.
- If the business also exports the cost of each item, the ABC classification should be run on margin as well as on revenue: an expensive tractor with a thin margin may matter less than a spare part with a high one.
- Machinery sales are likely to follow the agricultural calendar, so category trends should be read against planting and harvest seasons.

Limitations. Olist publishes neither product names nor costs, so products are described only by category and price, and the classification uses revenue rather than margin. The period covers 20 months, and the six-month rule for slow movers is a business convention, not a statistical result.

6. Question 2 · Demand forecast and inventory

We forecast the units sold per month from August to October 2018. Seasonal models such as Holt-Winters and SARIMA need two complete years to estimate a yearly pattern and there are 19 months, so the comparison included two simple baselines and the non-seasonal versions of both families. The model was chosen with a rule set before seeing the results: the lowest MAPE in a rolling evaluation with 15 forecasts.

Model	MAPE, rolling evaluation	MAPE, holdout	MAE, rolling evaluation (units)
3-month moving average	7.8%	7.8%	595
Naive (last month)	8.7%	8.1%	665
Holt (damped trend)	14.9%	28.6%	1,107
ARIMA (order chosen by AIC)	16.7%	10.5%	1,267

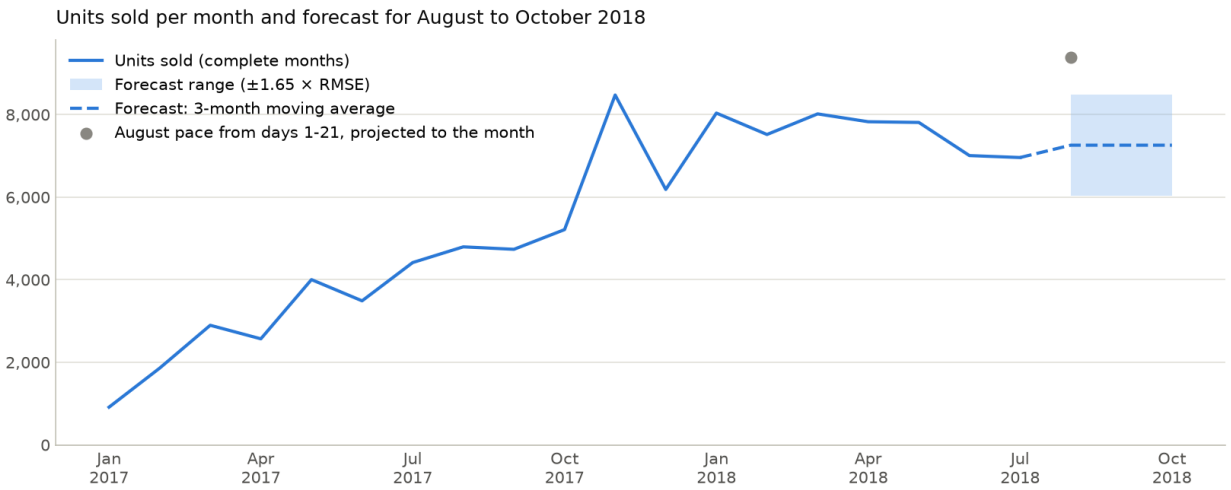


Figure 6. *Units sold per month and forecast for August to October 2018. The gray dot is the pace of August 1 to 21 projected to the month, 22.7% above the forecast.*

Suggested inventory by category

For the 18 class A categories, the suggested monthly stock is the forecast plus a safety stock of 1.65 times the root mean squared error of each category's past forecasts, which corresponds to a 95% service level.

Category	Monthly forecast	Safety stock	Suggested stock	Error (WAPE)
Health beauty	799	240	1,040	19.0%
Computers accessories	424	547	971	49.7%
Housewares	586	240	827	25.8%
Bed bath table	707	114	822	7.1%
Watches gifts	540	227	768	24.0%
Sports leisure	456	255	712	23.7%
Furniture decor	478	163	641	16.6%
Auto	315	96	411	15.2%
Toys	187	217	404	49.4%
Telephony	229	124	353	22.7%
Garden tools	216	135	352	26.4%
Stationery	162	162	325	48.6%
Baby	225	66	292	16.2%
Perfumery	194	60	255	15.7%
Cool stuff	131	120	252	32.5%
Pet shop	157	78	236	29.6%
Office furniture	87	104	191	42.2%
Computers	11	18	30	108.8%

Answer to question 2

Business question. How much will sell in the next 3 months, and how much inventory is needed?

Analysis. We compared two simple baselines (naive and 3-month moving average) with Holt and ARIMA on the 19 complete months of data, using a holdout and a rolling evaluation with 15 forecasts, and chose the model with the lowest MAPE. With that model we forecast total units and revenue for August to October 2018 and the demand of the 18 class A categories, adding a safety stock based on each category's past forecast errors.

Evidence.

- The 3-month moving average had the lowest error (7.8% MAPE), ahead of the naive forecast (8.7%), Holt (14.9%) and ARIMA (16.7%). Seasonal models could not be fitted: they need at least 24 months of history, and there are 19.
- Forecast: 7,259 units and about R$ 900,000 in revenue per month, with a range of 6,034 to 8,484 units.
- The first 21 days of August ran at 303 units per day. Projected to the month, that pace leaves the forecast 22.7% short: the models did not anticipate the jump.
- At category level the error rises to about 24%, ranging from 7.1% in Bed bath table to 108.8% in Computers.

Insight. With a short history, simple models forecast better than complex ones, because the 2017 growth misleads the trend models. The forecast is reliable for the overall level of demand, not for sudden jumps, and its quality varies a lot between categories: stable categories can be planned with it, while irregular or seasonal ones need more than an average.

Recommendation.

1. Plan monthly purchases with the 3-month moving average and recalculate it every month with the latest sales.
2. Keep a safety stock per category based on its own forecast error, as in the table above, and update it when the error changes.
3. For seasonal or irregular categories (Toys, Stationery, Computers accessories), adjust the plan with the business calendar, such as Christmas or back to school, instead of relying on the average.
4. Order expensive items with intermittent demand, such as Computers, on demand instead of keeping them in stock.
5. Compare actual sales with the forecast range every week. A month like August 2018, running above the upper end, calls for an early restock.

Application to the machinery business. The same process works with the business's data, with three adjustments:

- Machinery suppliers can take weeks or months to deliver, while this analysis assumed delivery within the month. The safety stock grows with the delivery time: with a lead time of L months, it is multiplied by the square root of L.

- Tractors and implements behave like Computers here: few units, high value and intermittent demand, which are better managed by order than by forecast. The forecast is most useful for spare parts, which sell more steadily.
- Agricultural sales follow the planting and harvest calendar. Once the business has at least two years of data, seasonal models such as Holt-Winters and SARIMA become viable and should be compared again with the same method.

Limitations. The history covers 19 months, without a complete yearly cycle; forecasts are monthly only; and the safety stock assumes that future errors will look like past ones. Olist is a marketplace where sellers hold the stock, so the inventory figures illustrate the method rather than an actual purchase plan.

7. Question 3 · Customer location and new branch

We located every order through the customer's zip code, measured each state's demand and the service it receives today, and ranked the 27 states with criteria defined before looking at the results: 50% demand (number of customers) and 50% service gap (average freight, median delivery days and share of late deliveries), each converted into a percentile.

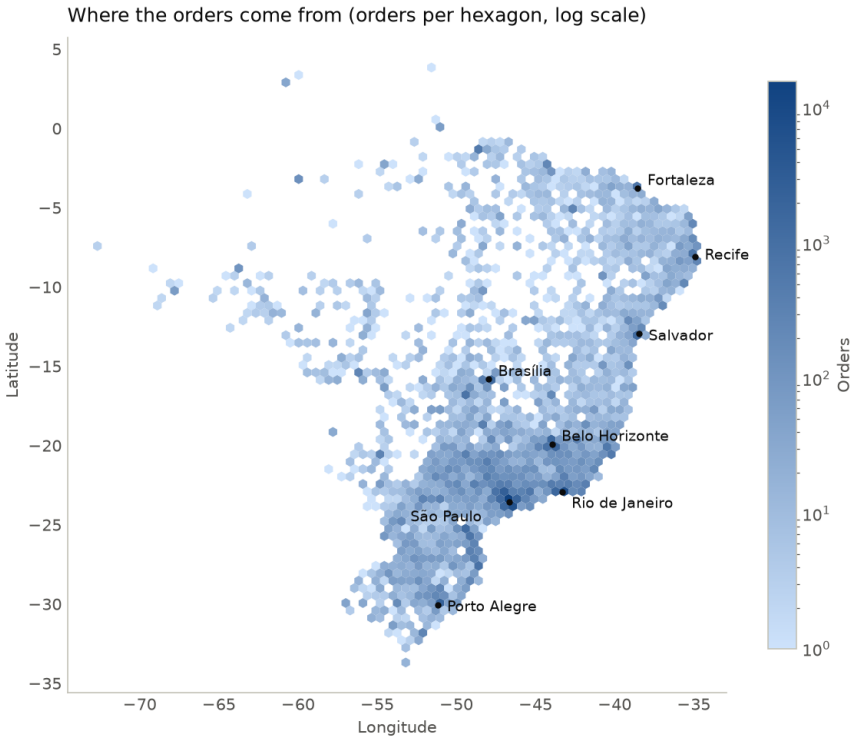


Figure 7. *Where the orders come from. The color scale is logarithmic because otherwise São Paulo would hide the rest of the country.*

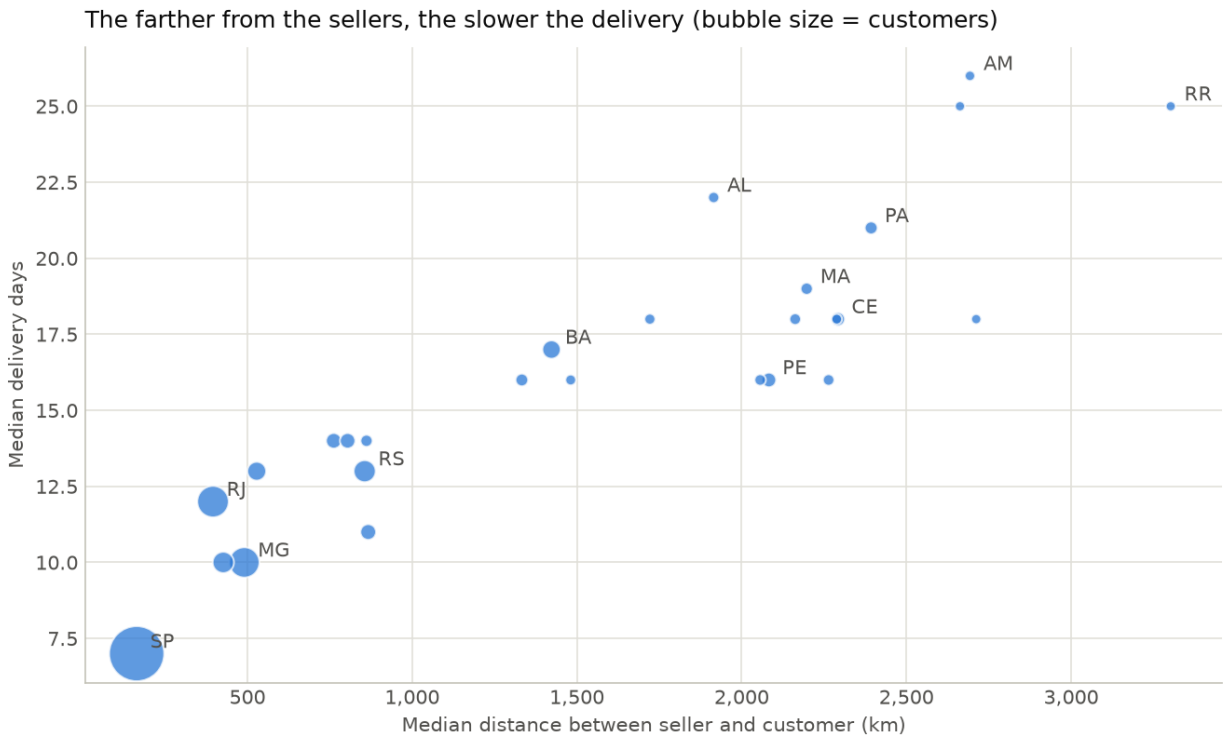


Figure 8. *Median distance between seller and customer against median delivery days, one point per state (rank correlation of 0.89).*

State	Region	Customers	Average freight	Median delivery days	Late orders	Score
Bahia (BA)	Northeast	3,155	R$ 29.97	17	12.2%	0.69
Maranhão (MA)	Northeast	696	R$ 42.78	19	17.5%	0.67
Rio de Janeiro (RJ)	Southeast	11,879	R$ 23.96	12	12.1%	0.67
Pará (PA)	North	918	R$ 39.66	21	11.3%	0.65
Ceará (CE)	Northeast	1,252	R$ 36.48	18	13.8%	0.64

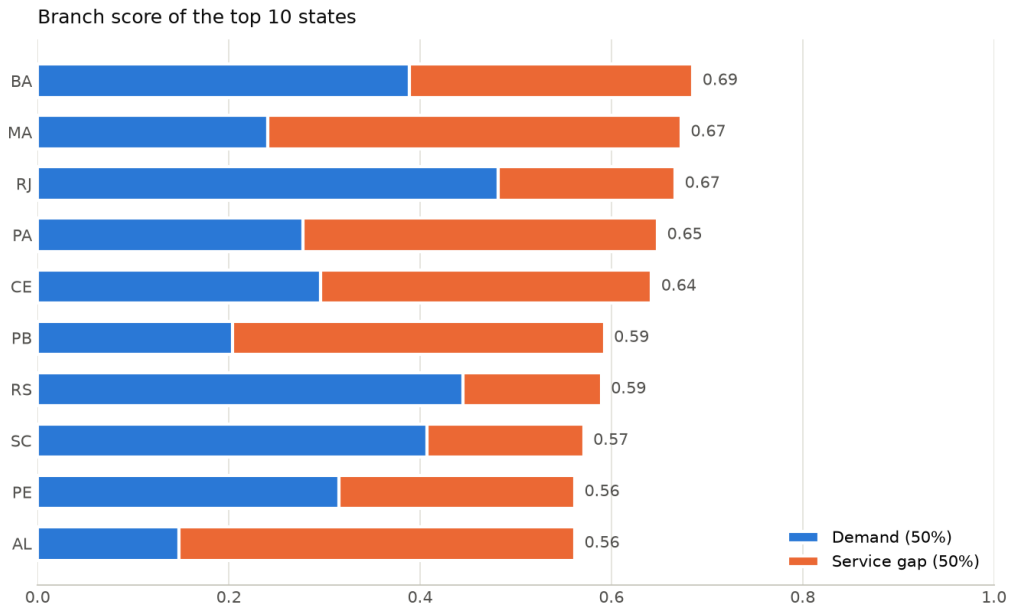


Figure 9. *Branch score of the top 10 states, split into demand and service gap.*

Answer to question 3

Business question. Where do customers come from, and where should a new branch open?

Analysis. We located every order through its zip code prefix, measured each state's demand and the service it receives today (freight, delivery time, late deliveries and distance to the sellers), and ranked the 27 states with criteria defined before looking at the results: 50% demand and 50% service gap. We tested the ranking with other weights and compared the main cities of the leading region by their average distance to the region's orders.

Evidence.

- Customers are concentrated in the Southeast: the state of São Paulo has 41.9% of them, and the city of São Paulo has more customers (14,500) than the whole Northeast (8,786).
- The farther from the sellers, the slower the delivery (rank correlation of 0.89 between distance and delivery days): 7 days in São Paulo, 17 in Bahia and 25 or more in Amazonas, Amapá and Roraima.
- Bahia leads the ranking (0.69) and is the most robust candidate when the weights change (average position 3.3). Three of the top five states are in the Northeast.
- Within the Northeast, Salvador is 576 km on average from the region's orders, close to the most central city (Aracaju, 520 km), and it has the region's largest customer base (1,154).

Insight. Demand and service quality pull in opposite directions: the largest markets are the best served, and the worst served markets are small. Bahia and the Northeast are where both criteria meet: a meaningful market that waits more than two weeks for its orders (17 days in Bahia) and pays about 70% more freight than São Paulo (R$ 30 against R$ 17). Rio de Janeiro ranks high because of its size, but its problem is late deliveries rather than distance, so a branch is not the obvious fix there.

Recommendation.

1. If a new branch is opened, Salvador is the best-supported location: it serves the largest market of the most promising region and sits close to the center of the Northeast's demand.

2. Before deciding, the business should state explicitly how much it values growing where customers already are (demand) against improving service where it is poor, because the ranking changes with that choice.
3. In Rio de Janeiro, investigate the causes of the late deliveries (carriers, delivery promises) before considering a branch.

Application to the machinery business. The same criteria translate directly to towns and rural areas. Demand becomes the number of customers or machines sold in each area, and the service gap becomes the distance and travel time from the current store, the cost of transport and the response time for spare parts and repairs. With the business's sales data, the geographic notebook ranks the areas in the same way, and the weights should be agreed with the owners before looking at the results. A branch also has costs (rent, staff, stock) that are not in the sales data, so the ranking produces a shortlist, not the final decision.

Limitations. Olist sellers ship from their own locations, so a branch is a simplification of how the marketplace works. The analysis leaves out the cost of opening a branch, the location of competitors and population data, and its ranking depends on the chosen weights. Distances are straight lines between zip code areas, not road distances.

8. Question 4 · Customers who do not come back

8.1 How churn is defined

A customer who does not come back can be described as one who never makes a second purchase. To build a predictive model that idea needs precision: from when the customer is observed, until when, what counts as coming back and what information the model may use. These were the decisions, each one supported by the data in notebook 06_customer_churn.

What counts as a purchase. Of the 2,801 customers with more than one order, 786 placed all of them on the same day: it is a single purchase split across several sellers, not a customer who came back. That is why a **purchase occasion** is defined as a day with at least one order. The customers who really come back are 2,015 (2.2%).

Prediction moment. The model predicts when the first purchase has been delivered and, if the customer rated it, reviewed. At that moment the delivery time, the delay and the review score are known, so the model can use them without looking into the future. That moment arrives a median of 11 days after the purchase.

A 180-day prediction window. Returning customers take a median of 74 days to come back: 54.9% return within 90 days, 75.9% within 180 and 95.8% within a year. A longer window captures more returns, but leaves fewer customers with a complete window before the end of the data:

Window	Customers with a complete window	Came back	Return rate	Last usable first purchase
90 days	71,538	757	1.06%	2018-05-20
180 days (chosen)	49,936	875	1.75%	2018-02-19
365 days	18,305	562	3.07%	2017-08-17

With 90 days, almost half of the customers who do come back (45.1%) would be labeled as lost just for taking more than three months. With 365 days, only customers who first bought before August 2017

could be used. The 180-day window captures three quarters of the returns and leaves the most returns to learn from (875).

Target and cohort. `returned = 1` if the customer makes a new purchase within 180 days after the prediction moment, and `churn = 1 − returned`. The cohort gathers the customers whose first purchase took place between January 5, 2017 and February 19, 2018, the only ones whose full window ends before August 21, 2018, the last day with complete data:

Step	Customers
All customers	93,350
First purchase from January 2017	93,086
Full 180-day window observed before the end of the data	50,105
Did not come back before the prediction moment	49,936

Of the 49,936 customers in the cohort, 875 came back (1.75%): the churn rate is 98.25%.

How it is built in SQL. The query `churn_first_purchase.sql` groups each customer's orders by day (purchase occasions), numbers them by date with `ROW_NUMBER()`, takes the first occasion with its features and joins it with the second one to know whether the customer came back and when. The prediction moment is the later of the delivery date and the review date.

How data leakage is avoided. The model only uses information about the first purchase known at the prediction moment: amount, units, category, freight, payment method, installments, delivery days, delay, review score and comment, region and month. The number of orders, total spend, recency and RFM scores are forbidden, because they are computed with later purchases and contain the answer. In addition, 169 customers who came back before the prediction moment were left out, and the notebook checks four conditions automatically before training:

- The prediction moment is on or after the delivery of the first purchase.
- The prediction moment is on or after the review, when there is one.
- Every return counted in the target happens after the prediction moment.
- Every 180-day window ends before the last day with complete data.

Temporal validation. The models are trained with the customers whose first purchase was before October 2017 (25,314 customers, of whom 1.98% come back) and evaluated with the later ones (24,622 customers, 1.51%), as they would be used in practice: learning from the past to predict the future.

8.2 Results

With this definition we compared return rates across groups, with 95% confidence intervals so that differences that could be due to chance are not read as real, and trained two models.

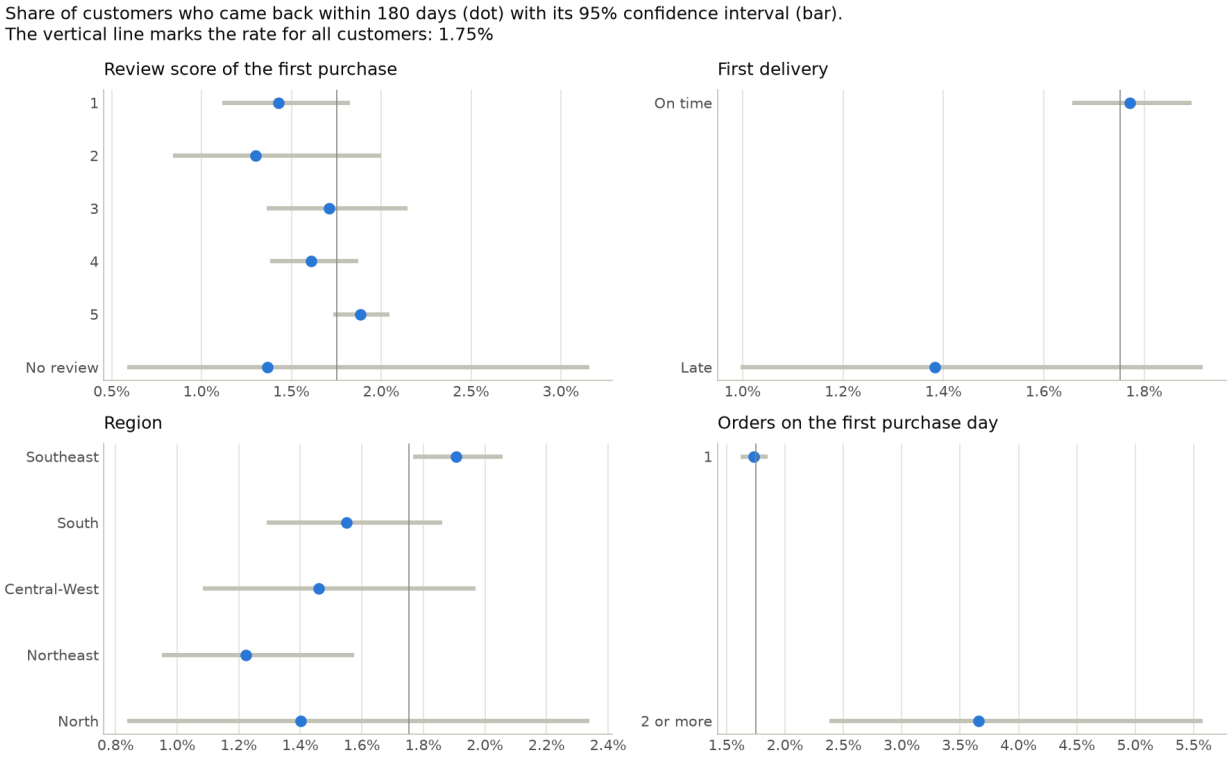


Figure 10. *Share of customers who came back within 180 days, with 95% confidence intervals. Region and first-day orders show clear differences; review score and late delivery do not.*

Model	ROC AUC	PR AUC	Base rate	Found in the top 10%	Precision in the top 10%
Random forest	0.571	0.019	1.51%	15.0%	2.27%
Logistic regression	0.557	0.018	1.51%	13.7%	2.07%

ROC AUC is the probability that a customer who came back gets a higher score than one who did not: 0.5 is chance and 1 a perfect ranking. Picking 10% of the customers at random finds 10% of those who come back.

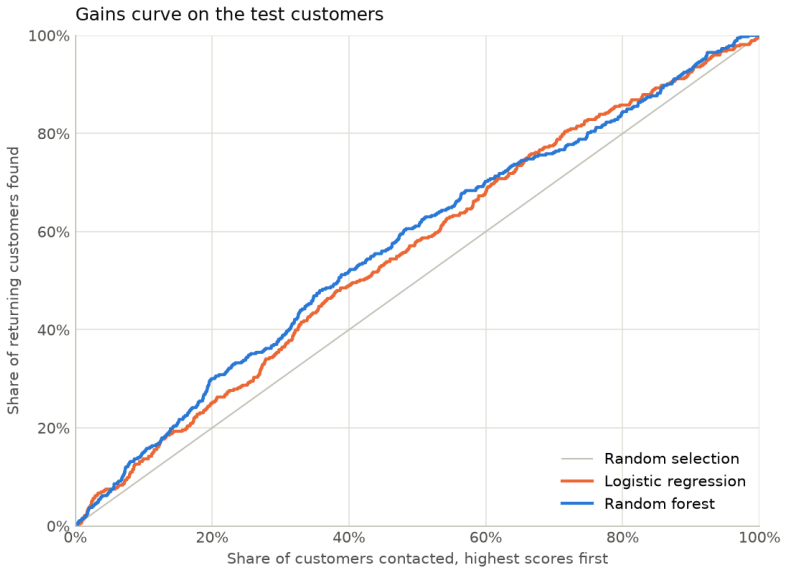


Figure 11. *Gains curve on the test customers: share of returning customers found when contacting a share of the customers, starting with the highest scores.*

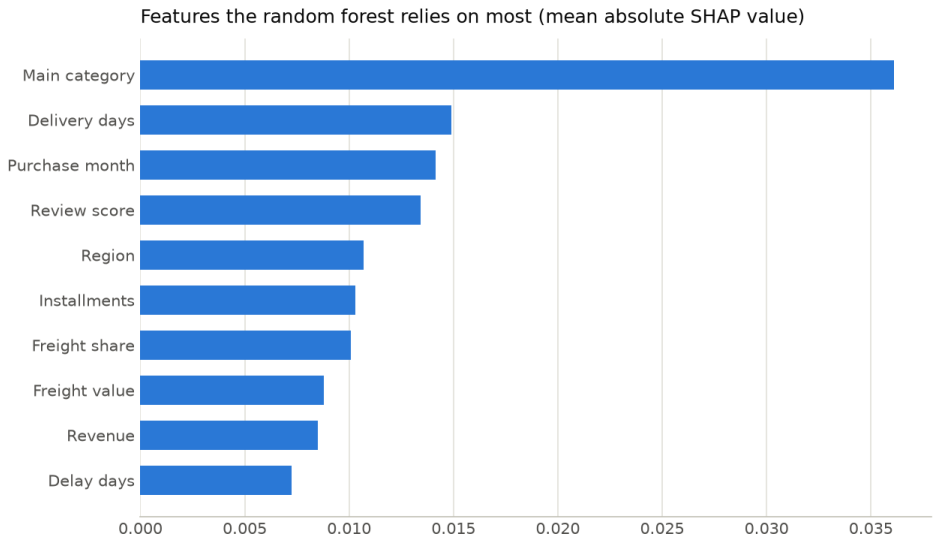


Figure 12. *Features the random forest relies on most. SHAP values describe the model's predictions, not the causes of churn.*

Answer to question 4

Business question. Why do some customers not come back, and who is about to leave?

Analysis. We defined churn as not making a new purchase within 180 days after the first purchase has been delivered and reviewed, built the dataset with SQL using only information available at that moment, and checked it against data leakage (section 8.1). We compared return rates across groups with 95% confidence intervals, trained a logistic regression and a random forest with a temporal split, explained the better model with SHAP and used it to score the 38,938 most recent customers.

Evidence.

- 97.8% of the customers bought on a single day, which is why RFM segmentation was not useful here.
- Of the 49,936 customers in the cohort, 875 (1.75%) came back within 180 days: churn is 98.25%.
- Clear differences: Southeast customers came back more often than Northeast customers (1.91% against 1.22%), customers with two or more orders on their first day came back twice as often (3.66% against 1.73%), and November and December buyers came back less often than August buyers (1.42% and 1.49% against 2.37%).
- Review score and late delivery point in the expected direction, but their differences are not statistically conclusive.
- The best model, a random forest, reaches a ROC AUC of 0.571: the 10% of customers with the highest scores contain 15% of those who came back.
- Among the recent customers, the group with the lowest return score had 30.6% late deliveries and an average review of 3.3, against 0.1% and 4.7 in the group with the highest score.

Insight. Almost nobody comes back, and the first purchase explains little of who does. What the data does show is that coming back is associated with a smooth experience (being close to the sellers, fast deliveries, good reviews) and with buying everyday products rather than gifts. The model is not strong enough to single out individual customers reliably, but it ranks them better than chance, and its risk groups are consistent with the delivery experience.

Recommendation.

1. Treat retention as something to build: with 98% churn, a second purchase will not happen on its own, so a post-purchase program (follow-up messages, offers on related everyday products) is needed.
2. Use the risk groups for low-cost actions, such as an email campaign, where a weak model is still useful, and not for expensive individual actions.
3. Prioritize delivery reliability in the regions far from the sellers, where customers come back least, and connect it with the branch decision of question 3.
4. Test every retention action with a control group (A/B test). These associations do not prove causes, and only an experiment shows whether an action really brings customers back.

Application to the machinery business. The definition carries over with two changes. First, the window has to match the purchase cycle: tractors and implements are bought only every few years, so churn should be measured on spare parts and service, with a window of 6 to 12 months as the proposal suggests. Second, the business probably has far more repeat purchases than Olist, which makes both RFM and the predictive model more useful there. The same safeguards apply: a prediction moment, features known at that moment and a temporal validation.

Limitations. The model predicts weakly (ROC AUC 0.571) and has only 875 returning customers to learn from. The data does not include marketing contact, prices at competitors or customer demographics, which probably explain much of the decision to come back. All the results are associations, not causes, and the 180-day window is a choice that shapes the target.

9. Power BI dashboard

The dashboard brings the four answers together, one page per business question. Each page shows the question as its title and the main insight as its subtitle. It reads the CSV files produced by the notebooks, so every figure matches this report, and it can be refreshed when the notebooks are run again with the business's data. The build guide in the repository (`docs/dashboard_guide.md`) explains how to rebuild it step by step.

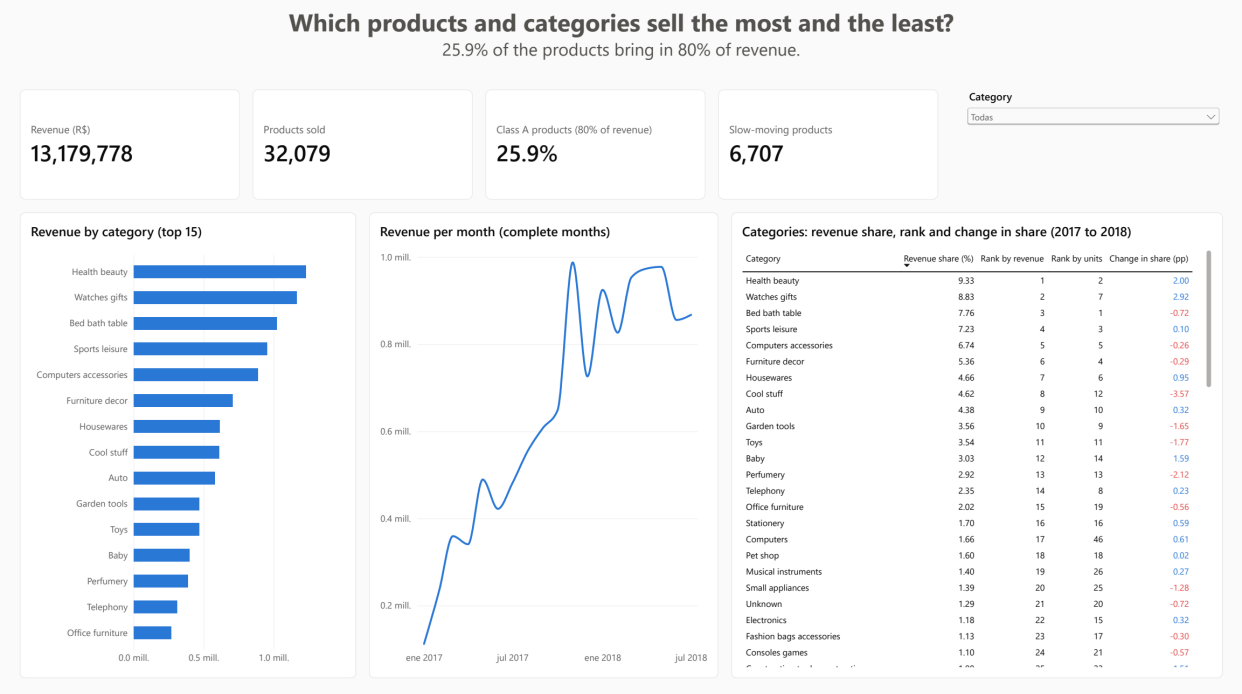


Figure 13. *Products page. Key figures, revenue by category, revenue per month and the change in each category's share of revenue (blue: gained, red: lost). The category filter updates the whole page.*

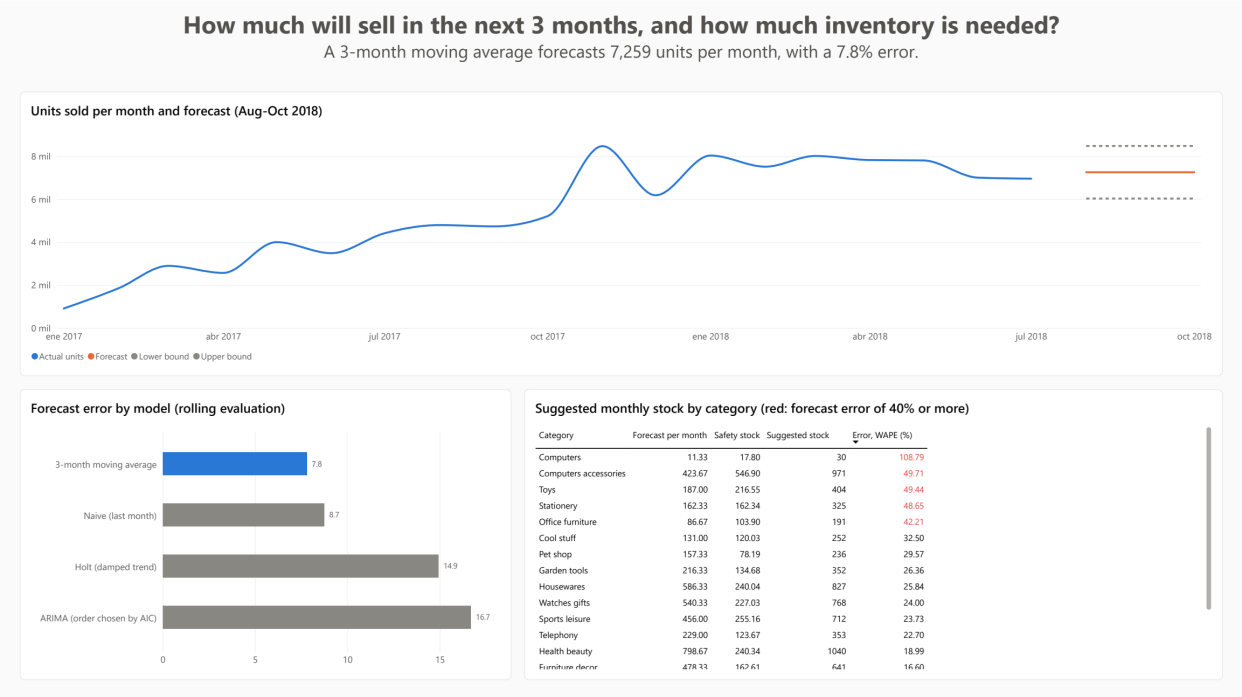


Figure 14. *Inventory page. Units sold per month with the forecast and its range, the forecast error of each model and the suggested stock by category, with unreliable forecasts in red.*

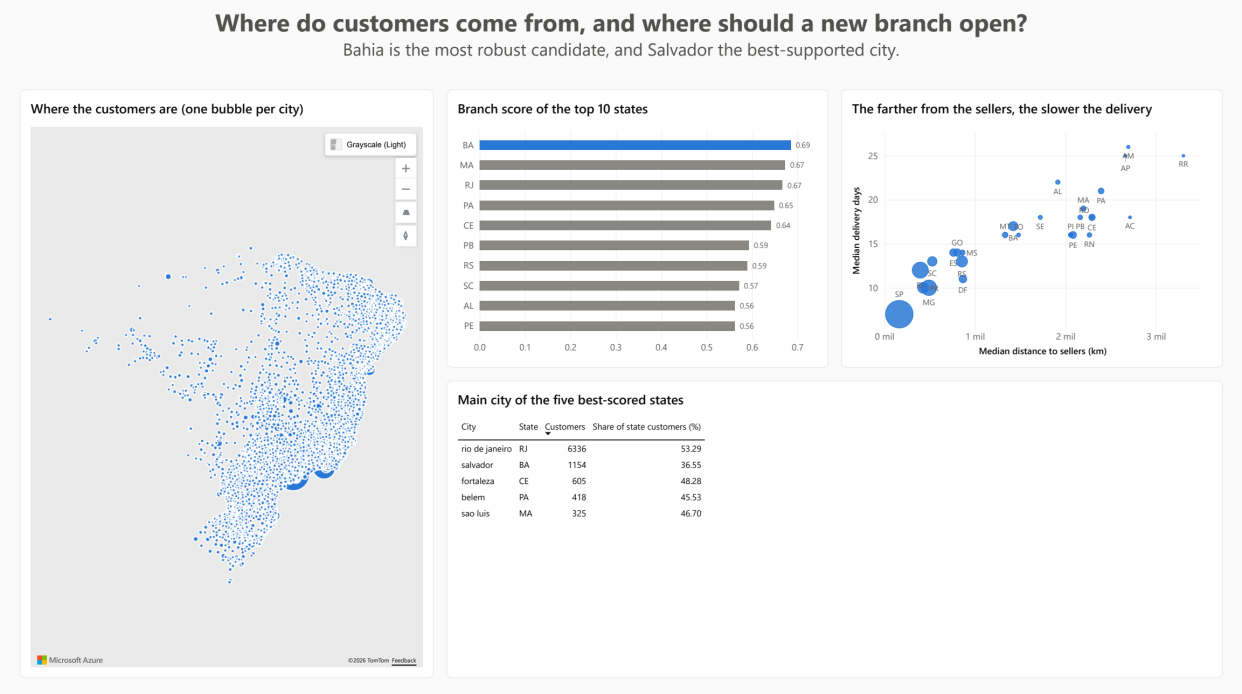


Figure 15. *Locations page. Customers by city, the branch score of the top 10 states, the relationship between distance and delivery days, and the main city of the five best-scored states.*

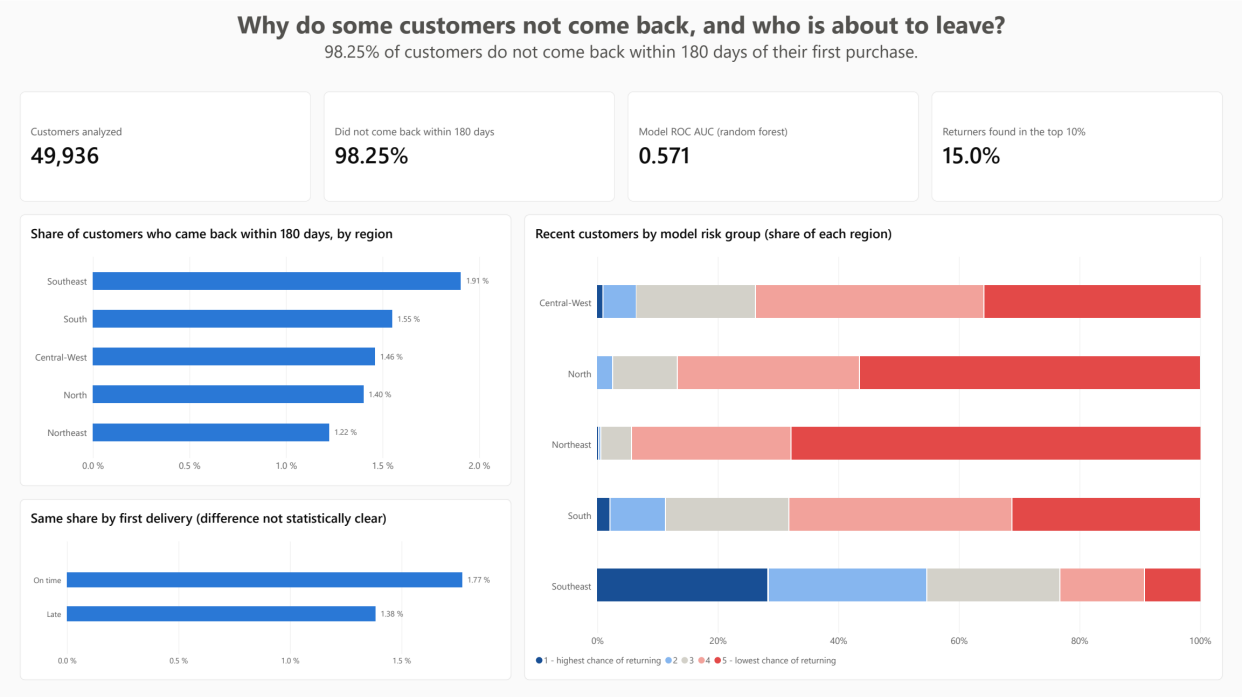


Figure 16. *Customers page. Cohort figures, return rate by region and by first delivery, and how the recent customers of each region split across the model's risk groups.*

10. Analytical decisions evaluated

Several decisions changed the results. Each one was evaluated with data before adopting it:

- **The end of the data is not a drop in sales.** Orders fell abruptly from August 22, 2018 in every order status, so August 2018 is treated as incomplete instead of being fed to the models.

- **Same-day orders are a single purchase.** 786 customers placed several orders on a single day; counting them as returning customers would have inflated the repeat purchase rate.
- **No seasonal forecasting models.** Holt-Winters and SARIMA need two complete years to estimate a yearly pattern. With 19 months the fit fails, and the notebook shows the error.
- **No hand-picked ARIMA order.** The order is chosen by AIC within each training window, because choosing it after seeing the test results would overstate its accuracy.
- **No RFM segmentation.** 97.8% of the customers bought on a single day, so frequency separates nobody.
- **Branch criteria set before the results.** The 50/50 weights were defined first and then tested with 70/30 and 30/70; Bahia stays among the top six in all three scenarios.
- **Confidence intervals in every group comparison.** With a 1.75% return rate, several differences that look important, such as review score or late deliveries, are not conclusive.

11. Limitations

- **Olist is a proxy.** It is a Brazilian marketplace where sellers ship from their own locations and hold their own stock, not an agricultural machinery business. The inventory and branch results illustrate the method, not an actual plan.
- **Short history.** 19 complete months, without a full yearly cycle, limit the forecast and any conclusion about seasons.
- **The churn definition shapes the results.** The 180-day window and the prediction moment are documented choices, and the model has only 875 returning customers to learn from.
- **Correlation is not causation.** Group comparisons and SHAP values show associations with coming back, not causes; only experiments can prove causes.
- **Missing information.** There are no product names, costs, margins, marketing contacts, competitor prices or customer demographics, and distances are straight lines between zip code areas.

12. Implications for the machinery business and next steps

- Run the ABC classification on margin as well as on revenue, and manage expensive items with intermittent demand, such as tractors, by order rather than by forecast.
- Plan spare parts with a simple forecast recalculated every month, and size the safety stock with the suppliers' actual lead times.
- Agree on the branch criteria and their weights before looking at the ranking, and add the cost of opening the branch to the decision.
- Measure churn on spare parts and service with a 6 to 12 month window, where repeat purchases are much more frequent than in Olist, and test retention actions with control groups.

The next step is to run the six notebooks again with the business's own data as soon as its sales system allows the export, following the application guide in the repository, and to refresh the dashboard with the new CSV files.

Annex · SQL queries

From Phase 2 on, every aggregation is a SQL query run on the SQLite database created in Phase 1. Each file contains one query:

Query	Purpose
`category_monthly.sql`	Units and revenue by category and month, with zeros in months without sales (recursive CTE)
`category_ranking.sql`	Category ranking by revenue and by units, with share and cumulative share
`churn_first_purchase.sql`	First purchase occasion of each customer and date of the second one, basis of the churn dataset
`city_summary.sql`	Customers, orders, revenue and approximate location of each city
`daily_sales.sql`	Orders and revenue by purchase day
`monthly_sales.sql`	Monthly indicators: orders, units, revenue, average ticket and customers
`order_locations.sql`	Coordinates of each order for the customer map
`order_values.sql`	Value and size of each order, for the ticket and units distributions
`orders_with_region_view.sql`	View that adds the Brazilian region to each order with a CASE expression
`payment_mix.sql`	Share of each main payment method
`product_abc.sql`	ABC classification of products with cumulative share (window functions)
`purchase_frequency.sql`	Orders and purchase occasions of each customer
`sales_by_hour.sql`	Share of orders and revenue by purchase hour
`sales_by_region.sql`	Sales, ticket, freight, late deliveries and median delivery days by region
`sales_by_state.sql`	Orders, revenue and share by state
`seller_location.sql`	Share of sellers located in São Paulo and in the Southeast
`shipping_coordinates.sql`	Customer and seller coordinates of each item, to measure shipping distances
`single_product_orders.sql`	Share of orders with a single distinct product
`state_summary.sql`	Demand and service indicators by state
`units_between.sql`	Units sold between two dates